

Biochemistry

1- Which vitamin deficiency is seen in infants with jaundice who undergo phototherapy?

C

A

B5

✓ B2

2- Which of the following fatty acid has 16 carbon atoms?

✓ Palmitic acid

Linoleic acid

Linolenic acid

Cerebronic acid

3- Binding of the prokaryotic RNA polymerase to promoter site is inhibited by?

Tetracycline

Puromycin

✓ Rifampicin

Streptomycin

4- Which of the following inactivates the 60S subunit of eukaryotic ribosomes?

Chloramphenicol

✓ Ricin

Diphtheria toxin

Cycloheximide

5- Which of the following mediates carrying genetic information from DNA to protein synthesis?

Sn-RNA

t-RNA

r-RNA

✓ m-RNA

6- Which of the following vitamins is involved in the structure of coenzyme A?

- ✓ Pantothenate (B₅)
- Niacin (B₃)
- Thiamine (B₁)
- Folate (B₉)

7- Which of the following vitamins is involved in gamma-carboxylation of glutamate?

- H
- C
- A
- ✓ K

8- Okazaki pieces are formed during the synthesis of?

- rRNA
- mRNA
- tRNA
- ✓ DNA

9- Where does a repressor bind to an operon?

- Catabolite activator site
- ✓ Operator
- Promotor
- Inducer

10- In vitamin B₆ deficiency which other vitamin deficiency is also observed?

- D
- C
- ✓ B₃
- B₁₂

11- A child with tall stature, loose joints, and detached retinas is found to have a mutation in type II collagen. Recall that collagen consists of a repeating tripeptide motif where the first amino acid of each tripeptide is the same. Which of the following amino acids is the recurring amino acid most likely to be altered in mutations that distort collagen molecules?

- ✓ Glycine
- Tyrosine
- Hydroxylysine
- Hydroxyproline

12- Lactose utilization by *E. coli* requires?

- ✓ Lac-Y
- Lac-Z
- Lac-I
- Lac-A

13- The oxygen carrier of muscle is the globular protein myoglobin. Which one of the following amino acids is highly likely to be localized within the interior of the molecule?

- Glutamic acid
- Arginine
- ✓ Valine
- Aspartic acid

14- Which of the following statements about Michaelis-Menten constant is correct?

- The K_m of an enzyme increases when the enzyme concentration is increased.
- A high K_m indicates a high affinity of an enzyme for its substrate.
- ✓ A low K_m indicates a high affinity of an enzyme for its substrate.
- The K_m of an enzyme is changed when the enzyme concentration is increased.

15- Actinomycin D is an inhibitor of?

- translation
- Replication
- DNA repair
- ✓ Transcription

16- What is the starting compound for the synthesis of sphingolipids?

Pyruvate

✓ Ceramide

lactate

Glucose

17- In which reaction does biotin participate?

Deamination

Decarboxylation

✓ Carboxylation

Amination

18- Which one of the following is the main form of vitamin D in plasma?

24,25(OH)₂-D₃

✓ 25-Hydroxycholecalciferol (25 (OH)-D₃)

Pre-vitamin D₃

1,25(OH)₂-D₃

19- What is the main function of t-RNA?

Proof reading

Transports messages to the cytoplasm

Inhibits protein synthesis

✓ Identifies amino acids and transports them to ribosomes

20- Which of the following mechanisms will remove uracil and incorporate the correct base?

Nucleotide excision repair

✓ Base excision repair

Mismatch repair

Direct repair

21- Which enzyme is used to join bits of DNA?

Endonuclease

✓ DNA ligase

Primase

DNA polymerase

22- What is the role of sigma factor in bacterial RNA polymerase?

Terminating RNA synthesis

Unwinding DNA template

Catalyzing RNA synthesis

✓ Positioning RNA polymerase correctly on the DNA template

23- According to Chargaff's rule, in a DNA molecule the percentage of?

$T+A=G+C$

$U+A=G+C$

$T+A=G+U$

✓ $T+G=A+C$

24- Which of the following vitamins is directly involved in the conversion of homocysteine to methionine?

C (Ascorbic acid)

✓ B12 (Cobalamin)

B6 (Pyridoxal)

K (Phyloquinine)

25- Which vitamin deficiency cause hemolytic anemia?

A

B3

H

✓ E

26- Which of the following is an incorrect statement about m-RNA?

Cap is added to the 5' end

Histone mRNAs lack 5' cap

✓ Exons are removed and introns are spliced together

Poly-A tail is added to the 3' end

27- The highest concentration of cystine can be found in?

Collagen

✓ Keratin

Chondroitin sulfate

Melanin

28- The phenomenon of mutarotation is due to which of the following?

Conversion of linear Glucose to circular form

Conversion of Glucose to Fructose

Conversion of D-Glucose to L-Glucose

✓ Conversion of α -D-Glucose to β -D-Glucose

29- Which of the following phospholipids is the main component of inner mitochondrial membrane?

Lecithin

Cephalin

✓ Cardiolipin

Plasmalogen

30- Which of the following statements about solutions of amino acids at physiological pH is true?

All amino acids contain positively charged side chains

Some amino acids contain only positive charges

✓ All amino acids contain both positive and negative charges

All amino acids contain negatively charged side chains

31- Which of the following is a deoxy sugar?

Glucosamine

Fructose

✓ L-Fucose

N-acetylneuraminic acid

32- Galactonic acid is caused by oxidation of which factor in galactose?

Type 1 alcohol (primary alcohol group)

Carbonyl group

Type 2 alcohol (secondary alcohol group)

✓ Aldehyde group

33- The enzyme responsible for the removal of supercoiling in replicating DNA, ahead of the replication fork, is

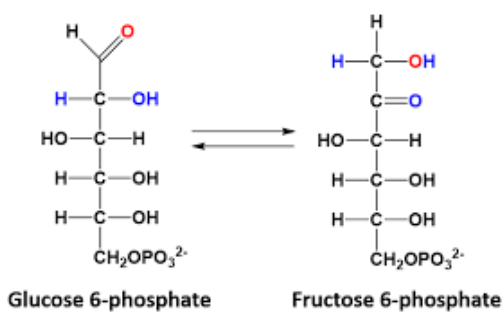
DNA polymerase

✓ Topoisomerase

Helicase

Primase

34- The following reaction is catalyzed by...?



✓ Isomerase

Oxidoreductase

Ligases

Hydrolases

35- When $S=3K_m$, the V is equal to..... ?

3 V_{max}



$\frac{3}{4} V_{max}$

4 V_{max}

$\frac{4}{3} V_{max}$

36- Which membrane fatty acids are precursors for eicosanoids?

C20 and C22 saturated fatty acids

C18 and C22 polyunsaturated fatty acids

C20 and C22 monounsaturated fatty acids



C20 and C22 polyunsaturated fatty acid

37- Which amino acid is involved in N-glycosidic bond formation?

Glycine

Tyrosine



Asparagine

Glutamate

38- Which of the following would produce Methemoglobinemia?



Oxidation of heme groups

Deletion of one β -globin locus

Deletion of one α -globin locus

Altered RNA processing at both β -globin loci

39- Which of the following lipoproteins is produced via the exogenous lipoprotein pathway?

High density lipoproteins

Low density lipoproteins



Chylomicrons

Very low density lipoproteins

40- All of these polysaccharides are homopolysaccharide except?

Starch

Inulin

Glycogen

✓ Heparin

41- Which of the following is an essential fatty acid?

Palmitic

✓ Linolenic

Oleic

Stearic

42- Which of the following reactions is required for proofreading, during DNA replication, by DNA polymerase III?

5' to 3' endonuclease activity

5' to 3' exonuclease activity

3' to 5' endonuclease activity

✓ 3' to 5' exonuclease activity

43- During synthesis of mature collagen fiber, which one of the following steps would occur within the fibroblast?

Formation of a triple helix

✓ Formation of covalent cross-links between molecules

Glycosylation of proline residues

Hydrolysis of procollagen to form collagen

44- The effect of reversible competitive inhibitor can be canceled out by?

Increasing the product concentration

✓ Increasing the substrate concentration

Increasing the pH

Increasing the temperature

45- Which vitamin has thiazole and pyrimidine ring in its structure?

- ✓ B1 (Thiamine)
- B3 (Niacin)
- B5 (Pantothenic acid)
- H (Biotin)

46- Which of the following leads to disruption of nucleosome structure?

- ✓ Acetylation
- Carboxylation
- Oxidation
- Phosphorylation

47- Which is an example of RNA-dependent DNA polymerase?

- ✓ Reverse transcriptase
- RNA polymerase I
- RNA polymerase II
- DNA ligase

48- The enzyme which hydrolyses Galactose -Glucose linkage is?

- Protease
- Amylase
- ✓ Lactase
- Maltase

49- Ibuprofen is a reversible inhibitor of the COX-1 enzyme which is involved in Arachidonic Acid metabolism. Blocking of enzyme action by blocking its active site is called as?

- Non-competitive inhibition
- ✓ Competitive inhibition
- Feedback inhibition
- Allosteric inhibition

50- Which option is correct for Cellobiose?

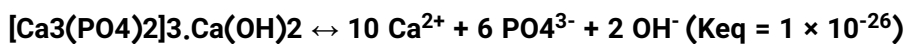
It has $\alpha 1 \rightarrow 4$ glycosidic bond

✓ It has $\beta 1 \rightarrow 4$ glycosidic bond

It contains Galactose and Glucose

It contains Glucose and Galactose

51- The organic component of bone is Hydroxyapatite. In blood ($\text{pH} \cong 7.4$) Hydroxyapatite dissociates as follows:



If K_{eq} for this reaction is 1×10^{-26} , then which of the following can correctly describe Hydroxyapatite?

✓ It is relatively insoluble under physiological condition

It dissolves only in basic solutions

It would dissolve well in calcium phosphate solution

It dissolves well under physiologic condition

52- What units make up Lactose with what bonds?

α -glucose+ α -glucose, $\alpha 1 \rightarrow 1$

β -glucose+ β -fructose, $\beta 1 \rightarrow 6$

✓ β -galactose+glucose, $\beta 1 \rightarrow 4$

α -glucose+ β -fructose, $\alpha 1 \rightarrow 2$

53- Which vitamin is involved in the synthesis of collagen protein?

B1

✓ C

D

K

54- How many spatial isomers does the linear form of aldohexoses have?

32

3

✓ 16

8

55- An example of glycerophospholipid involved in cell signaling is?

Phosphatidic acid



Phosphatidylinositol

Cardiolipin

Phosphatidylcholine

56- The backbone of phospholipids is?

sn-glycerol 1-phosphate

L-glycerol 1-phosphate



L-glycerol 3-phosphate

D-glycerol 3-phosphate

57- Blood is drawn from a child with severe anemia and the hemoglobin protein is degraded for the analysis of peptide and amino acids. Of the results below, which change in hemoglobin primary structure is most likely to correlate with the clinical phenotype of anemia?

Ile-Leu-Val to Ile-Ile-Val

Gly-Ile-Gly to Gly-Val-Gly

Gly-Asp-Gly to Gly-Glu-Gly



Leu-Glu-Ile to Leu-Val-Ile

histology

58- Which of the following is not RER function?



Synthesis of fatty acids

Synthesis of mitochondrial membrane proteins

Synthesis of membrane proteins

Synthesis of lysosomal enzymes

59- Which of the following has a 9×3 microtubules composition?

Flagella

Cilia



Centriole

Stereosilia

60- Which of the following is not present in the RER membrane during the protein synthesis process?

Signal-Recognition Particle receptor

Translocator complex

Ribosome receptor



Signal peptide receptor

61- Lipid Raft structure located in which part of the cell membrane?



Integral proteins sides

Peripheral proteins sides

Cytoplasmic side

Extracellular side

62- All of the following are true about apoptosis except:

In apoptosis, intracellular proteases are activated

In apoptosis, apoptotic bodies are formed

Apoptosis occurs in fetal development



Apoptotic cells resulted in inflammation

63- Genetic disorders of intermediate filaments is most observed in which of the following structures?

Cell Membrane



Nucleous

Proteasome

Glycogen

64- Which of the following cell organs usually has polarity?

Ribosome



Golgi apparatus

Lysozyme

Peroxisome

65- Which of the following terms is applied to the smallest chromosome subunit in eukaryotic cells?

Chromosome

✓ Nucleosome

heterochromatin

Euchromatin

66- In which of the following cases the protein not related to the relevant organelle?

Peroxisome-catalase

Mitochondria-Purine

Proteasome - Ubiquitin

✓ RER - Clatrin

Cell Physiology

67- Which of the following has "electrogenic nature"?

Na-glucose transporter

Ca transporter

✓ Na-K pump

Na-H exchanger

68- The recognition of chemical messengers by cells typically begins by

activating some enzymes

✓ interaction with a receptor at that cell

activating second messenger systems

gating of ion channels

69- The most abundant substance in our body is

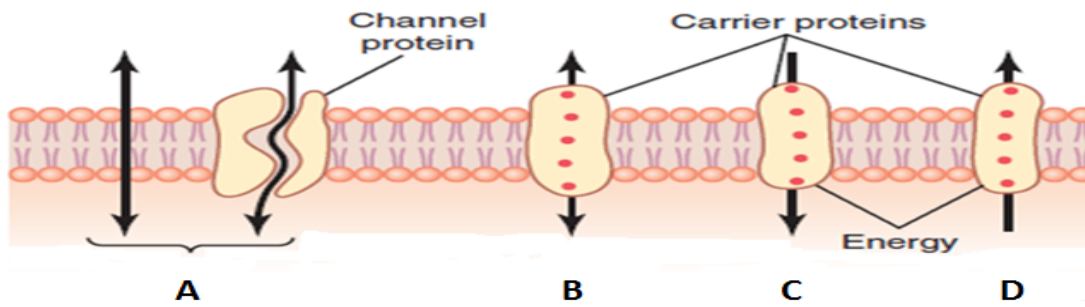
✓ water

protein

fat

carbohydrate

70- Which of the following shows "simple diffusion"?



D

B

C

A



71- In this type of communication, the products of cells diffuse in the ECF to affect neighboring cells that may be some distance away

Autocrine

Paracrine

Endocrine

Neural



72- Which type of glucose transporters are involved in "basal glucose uptake"?

GLUT 6

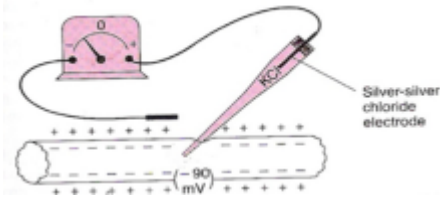
GLUT 4

GLUT 5

GLUT 1



73-



What does this figure show?

Measurement of the cytoplasmic resistance

✓ Measurement of the membrane potential

Measurement of the membrane selectivity

Measurement of the membrane resistance

74- Depolarization stage of action potential is caused as a result of

✓ opening of voltage gated Na channels

closure of voltage gated K channels

opening of voltage gated K channels

closure of voltage gated Na channels

75- What happens when a hormone or neurotransmitter is present in excess?

✓ The number of active receptors generally decreases.

Up-regulation process occurs.

Metabotropic receptors get activated.

Nothing happens because receptors are not static components of the cell.

76- In which second messenger formation, "phospholipase C" is involved?

Cyclic GMP

Ca ions

Cyclic AMP

✓ Inositol trisphosphate & diacylglycerol

77- How much is the membrane potential when the membrane has no permeability to X ions.

[X] inside: 5 [X] outside: 50 Log 10: 1 Log 0.1: -1

✓ X ions have no effect on determination of membrane potential.

+50 mv

+61 mv

-61 mv

78- Which type of channels contribute in both depolarization and repolarization phases of an action potential?

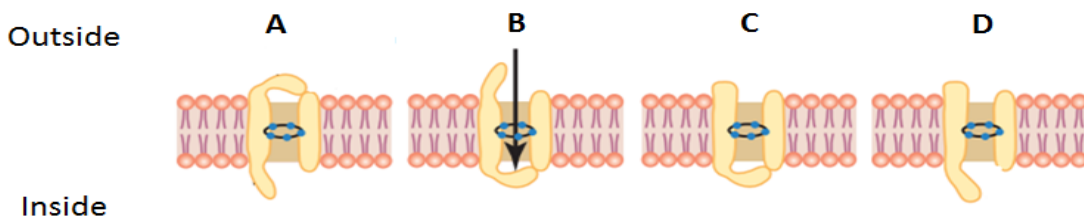
Voltage gated Ca channels

Every voltage gated channel is involved just in one phase of action potential.

Voltage gated K channels

✓ Voltage gated Na channels

79- Which of the following shows the inactivated state of voltage gated Na channels?



A

D

✓ B

C

80- Which of the following is considered to be a positive feedback?

Plasma glucose regulation

Blood pH regulation

Arterial blood pressure regulation

✓ Normal vaginal delivery